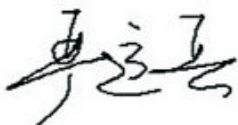
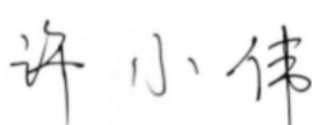
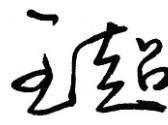


| 院教学指导委员会主任<br>(院长)                                                                | 学院分管教学<br>(副院长)                                                                   | 审核人<br>(专业责任教授团队负责人)                                                               | 执笔人                                                                                 |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |

## 车辆工程专业“战略性新兴产业（支柱）产业”培养方案

### Curriculum for Undergraduate of Automotive Engineering Major “Talents of Strategic and Emerging (Pillar) Industry”

#### 一、专业介绍

本专业起源于1972年开办的矿用汽车专业，2003年车辆工程专业开始招生，现为国家级一流本科专业。该专业旨在培养能在整车及零部件相关领域从事设计、研发、制造、集成及测试等工作的卓越应用型人才。2007年获批国家级专业实验教学示范中心，2009年获批湖北省重点学科，2011年车辆工程专业实施湖北省首批战略性新兴产业人才培养计划项目，2012年成为中国汽车工程学会会员单位，2016年获批教育部产教融合促进计划试点，2019年获批湖北省一流本科专业建设点，2021年获批国家级一流本科专业建设点。

#### I. Professional Introduction

This major originated from the mining automobile major established in 1972, and the vehicle engineering major began to enroll students in 2003. Now it is a national first-class undergraduate major. This major aims to train outstanding applied talents who can be engaged in design, research and development, manufacturing, integration and testing in the field of vehicle and parts. In 2007, it was approved as a national professional experimental teaching demonstration center; in 2009, it was approved as a key discipline in Hubei Province; in 2011, it was implemented as the First Batch of Strategic Emerging Industries Talent Cultivation Program in Hubei Province; in 2012, it became a member of the China Society of Automotive Engineering; in 2016, it was approved as a pilot program of the Ministry of Education to promote the integration of industry and education; in 2019, it was approved as a first-class undergraduate major construction point in Hubei Province; in 2021, it was approved as a national first-class undergraduate major construction point.

#### 二、培养目标

本专业培养适应社会经济发展需要和行业发展需求，德、智、体、美、劳全面发展的社会主义事业合格建设者和可靠接班人；培养具备扎实的自然科学基础理论及融合机械、信息和计算机的车辆工程专门知识，具有较强工程实践能力、一定创新创业精神和能力及国际视野，能在新能源及智能汽车相关领域从事设计、制造、集成及测试等工作的卓越应用型人才，期待学生毕业五年左右达到以下目标：

1. 具有良好的人文社会科学素养、工程职业道德和社会责任感；
2. 能综合运用信息、计算机和车辆工程多学科知识、原理和现代工具从事汽车产品设计、生产制造、系统集成及测试实验等工作；

3. 熟悉车辆工程领域行业规范和法律法规并主动遵守，能在工程实践中充分考虑社会、环境、法律、安全、健康及文化等因素；
4. 能够进行跨文化和跨领域的团队协作和沟通交流，具备多学科团队的组织与协调能力；
5. 具备可持续发展理念、国际视野及创新创业精神和能力；
6. 能够通过不断学习适应汽车科技和汽车产业转型发展要求。

## II. Training objectives

This major trains qualified builders and reliable successors of the socialist cause who meet the needs of social and economic development and automobile industry development, and develop comprehensively in moral, intellectual, physical, aesthetic, and labor; The students should be equipped with a solid basic theory of natural science and vehicle engineering expertise that integrates machinery, information and computers, and should have strong engineering practice ability, innovation & entrepreneurship, and an international perspective, and should be a high-quality applied talents who can engage in vehicle product design, manufacture, system integration, test and experiment in the field of intelligent vehicle. The graduates should achieve the following goals in about five years:

1. Possess good humanities and social science literacy, engineering professional ethics and a sense of social responsibility;
2. Be able to comprehensively use multidisciplinary knowledge, principles and modern tools of information, computer and vehicle engineering to engaged in automotive product design, manufacturing, system integration, test experiments, etc.;
3. Familiar with industry norms and laws and regulations in the field of vehicle engineering and actively comply with them, and be able to fully consider social, environmental, legal, safety, health and cultural factors in engineering practice;
4. Be able to conduct cross-cultural and cross-field team collaboration and communication, and have the ability to organize and coordinate multidisciplinary teams;
5. Possess the concept of sustainable development, international vision, innovation and entrepreneurial spirit and ability;
6. Be able to adapt to the requirements of automotive technology and the transformation and development of the automobile industry through continuous learning.

## 三、专业特色

本专业秉承“强基础、重融合、突实践、求创新”的理念，植根湖北，辐射长江经济带，面向全国，坚持立德树人，培养适应社会经济发展需要和汽车行业转型升级需求的卓越人才。以新能源汽车产业发展的新需求为导向，秉承多学科交叉融合的理念，大力推进产学研协同育人，实施“科研促教、竞赛促学、小班授课”的教学模式。近年来，为新能源汽车整车和零部件等企业培养了大批优秀人才，形成了专业与企业间双向支撑，具有良好的产业特色和发展基础。

## III. Professional characteristics

Adhering to the concept of “strong foundation, integration, practice and innovation”, this major is rooted in Hubei, radiating to the Yangtze River Economic Zone and facing the whole country, adhering to the principle of “cultivating people with moral integrity”, and cultivating outstanding talents who can adapt to the needs of social and economic development as well as the needs of transformation and upgrading of the automobile industry. Guided by the new needs of new energy automobile industry development and adhering to the concept of multidisciplinary cross-fertilization, we vigorously promote the collaborative education of industry, academia and

research, and implement the teaching mode of “scientific research promotes teaching, competition promotes learning, and small class teaching”. In recent years, it has cultivated a large number of excellent talents for new energy vehicle and parts enterprises, formed a two-way support between the specialty and enterprises, and has good industrial characteristics and development foundation.

#### 四、毕业要求

| 毕业要求                                                                                           | 指标点                                                          |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1. 工程知识：能够将数学、自然科学、工程基础和专业用于解决车辆工程领域的复杂工程问题。                                                   | 1.1 能够运用数学、物理、化学及机械学科的符号、图形和文字等语言工具表述车辆工程领域的工程问题。            |
|                                                                                                | 1.2 能针对汽车的结构、理论和设计等工程问题建立数学模型并求解。                            |
|                                                                                                | 1.3 能够将数学、自然科学及机械学科等相关知识和数学模型方法用于推演、分析车辆工程领域的复杂工程问题。         |
|                                                                                                | 1.4 能够将数学、自然科学及机械学科等相关知识和数学模型方法用于车辆工程领域复杂工程问题解决方案的比较与综合。     |
| 2. 问题分析：能够应用数学、自然科学和工程科学的基本原理，识别、表达、并通过文献研究分析复杂车辆工程问题，以获得有效结论。                                 | 2.1 能够运用数学、自然科学和机械科学的基本原理，识别和判断车辆工程领域复杂工程问题的重要环节以及关键要素。      |
|                                                                                                | 2.2 能够运用数学、自然科学和机械科学的基本原理和数学模型方法正确表达汽车设计、制造、测试和集成等方面的复杂工程问题。 |
|                                                                                                | 2.3 能够认识到解决车辆工程领域复杂工程问题有不同方案，会通过文献研究寻求可替代的解决方案。              |
|                                                                                                | 2.4 能够运用工程科学的基本原理，借助文献研究，分析汽车设计、制造、测试和集成过程的影响因素获得有效结论。       |
| 3. 设计/开发解决方案：能够设计针对车辆工程领域复杂工程问题的解决方案，设计满足特定需求的汽车总成和零部件，并能够在设计环节中体现创新意识，考虑社会健康、安全、法律、文化以及环境等因素。 | 3.1 掌握汽车工程设计和产品开发全周期、全流程的基本设计 / 开发方法和技术，了解影响设计目标和技术方案的各种因素。  |
|                                                                                                | 3.2 能够针对整车性能需求，进行汽车零部件的设计。                                   |
|                                                                                                | 3.3 能够进行汽车总体及系统设计，在设计中体现创新意识。                                |
|                                                                                                | 3.4 在汽车总体、系统及零部件设计中能够考虑安全、健康、法律、文化及环境等制约因素。                  |
| 4. 研究：能够基于科学原理并采用科学方法对车辆工程领域复杂工程问题进行研究，包括设计实验、分析与解释数据，并通过信息综合得到合理有效的结论。                        | 4.1 能够基于科学原理，通过文献研究或相关方法，调研和分析车辆工程领域复杂工程问题的解决方案。             |
|                                                                                                | 4.2 针对复杂工程问题的解决方案，能够根据不同对象及其特征，选择研究路线，设计实验方案。                |
|                                                                                                | 4.3 能够根据实验方案构建实验系统，通过规范的操作，安全地开展实验，正确地采集实验数据。                |
|                                                                                                | 4.4 能够整理和分析实验数据、进行结果解释和总结，并综合各方面的信息归纳得到合理有效的结论。              |

| 毕业要求                                                                                             | 指标点                                                                                |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 5. 使用现代工具：能够针对车工程领域复杂工程问题，开发、选择与使用恰当的技术、资源、现代工程工具和信息工具，包括对复杂工程问题的预测与模拟，并能够理解其局限性。                | 5.1 了解车辆工程专业常用的现代仪器与设备、信息技术工具、工程工具和模拟软件的使用原理、功能和方法，并理解其局限性。                        |
|                                                                                                  | 5.2 能够选择与使用恰当的仪器、信息资源、工程工具和专业模拟软件，对车辆工程领域复杂工程问题进行分析、计算与设计。                         |
|                                                                                                  | 5.3 能够针对车辆工程领域复杂工程问题，选用满足特定需求的现代工具进行模拟和预测，并能够分析其局限性。                               |
| 6. 工程与社会：能够基于工程相关背景知识进行合理分析，评价车辆工程专业工程实践和复杂工程问题解决方案对社会、健康、安全、法律以及文化的影响，并理解应承担的责任。                | 6.1 了解车辆工程专业相关领域的技术标准体系、知识产权、产业政策和法律法规，理解不同社会文化对工程活动的影响。                           |
|                                                                                                  | 6.2 能分析和评价车辆工程专业工程项目及实践对社会、健康、安全、法律、文化的影响，以及这些制约因素对项目的影响，并理解应承担的责任。                |
| 7. 环境和可持续发展：能够理解和评价针对复杂车辆工程问题的专业工程实践对环境、社会可持续发展的影响。                                              | 7.1 知晓和理解环境保护和可持续发展的理念和内涵，了解车辆工程专业实践与环境及社会可持续发展之间的联系。                              |
|                                                                                                  | 7.2 能够站在环境保护和可持续发展的角度思考车辆工程专业工程实践的可持续性，评价汽车产品周期中可能对人类和环境造成的损害和隐患。                  |
| 8. 职业规范：具有人文社会科学素养、社会责任感，能够在工程实践中理解并遵守工程职业道德和规范，履行责任。                                            | 8.1 了解中国国情，树立和践行社会主义核心价值观，理解个人与社会的关系，明确个人作为社会主义建设者和接班人所肩负的责任和使命。                   |
|                                                                                                  | 8.2 理解诚实公正、诚信守则的工程职业道德和规范，具有创新创业精神，并能在车辆工程实践中自觉遵守。                                 |
|                                                                                                  | 8.3 理解工程师对公众的安全、健康和福祉，以及环境保护的社会责任，能够在车辆工程实践中自觉履行责任。                                |
| 9. 个人和团队：能够在多学科背景下的团队中承担个体、团队成员以及负责人的角色。                                                         | 9.1 能在多学科背景下的团队中与其他成员有效沟通，合作共事。                                                    |
|                                                                                                  | 9.2 能够在团队中独立或合作开展工作，具备组织、协调和指挥团队的能力。                                               |
| 10. 沟通：能够就车辆工程领域复杂工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令。具备一定的国际视野，能够在跨文化背景下进行沟通和交流。 | 10.1 能够就车辆工程领域的专业问题以图表、文稿和口头陈述等形式，准确表达自己的观点，与业界同行和社会公众进行包容性的沟通、交流和讨论。              |
|                                                                                                  | 10.2 具有一定的国际视野，了解车辆工程专业领域的国际发展趋势、研究热点，理解和尊重世界不同文化的差异性和多样性。                         |
|                                                                                                  | 10.3 具备跨文化交流的语言和书面表达能力，能就车辆工程专业问题，在跨文化背景下进行基本沟通和交流。                                |
| 11. 项目管理：理解并掌握工程管理原理与经济决策方法，并能在车辆工程领域多学科环境中应用。                                                   | 11.1 掌握汽车产品工程项目中涉及的管理和经济决策方法。                                                      |
|                                                                                                  | 11.2 了解汽车工程及产品全周期、全流程的成本构成，理解其中涉及的工程管理与经济决策问题，并能在多学科环境下设计开发解决方案的过程中，运用工程管理与经济决策方法。 |
| 12. 终身学习：具有自主学习和终身学习的意识，有不断学习和适应发展的能力。                                                           | 12.1 能认识到经济、社会 and 科技的发展进步对个人知识更新和能力提升的要求，具备自主和终身学习的意识。                            |
|                                                                                                  | 12.2 具有自主学习的能力，包括对技术问题的理解能力，归纳总结能力和提出问题能力等。                                        |

#### IV. Requirements

1. Engineering knowledge: Have ability of applying math, nature sciences, engineering foundations and professional knowledge to solve complicated engineering problems in the field of vehicle engineering;
2. Problem analysis: Have ability of applying basic theories of math, nature sciences and engineering sciences to identify, express complicated engineering problems in the field of vehicle engineering and analyze them through literature information to obtain effective conclusions;
3. Design / development solutions: Have ability of designing solutions for complicated engineering problems in the field of vehicle engineering and designing systems, parts and components to meet special requirements with innovative consciousness, considering society, health, safety, legislation, culture, environment etc.;
4. Research: Have ability of using scientific principles and approaches to study complicated engineering problems in the field of vehicle engineering including designing experiment, analyzing and interpreting data, to obtain reasonable conclusions through information integration;
5. Application of modern tools: Aiming at complicated engineering problems in the field of vehicle engineering, have ability to develop and chose suitable technology, resources, modern engineering tools and information technology tools to predict and simulate them, understanding their limitation;
6. Engineering and society: Based on the relevant engineering background knowledge, have ability to perform reasonable analysis and evaluate the influences of solutions of engineering practices and complicated engineering problems in vehicle engineering on society, health, safety, legislation, culture etc., understanding corresponding responsibilities;
7. Environment and sustainable development: Have ability to understand and evaluate the influences of engineering practice aiming at complicated engineering problems in the field of vehicle engineering on environment and sustainability;
8. Professional ethics and norms: With humanistic quality, sense of social responsibility, have ability of understanding and obeying professional ethics and norms and performing duties;
9. Individual and team: Have ability to act as individual, the team member or the leader in the team under a multidisciplinary background;
10. Communication: Have ability to communicate with the general public and peers on complicated engineering problems in the field of vehicle engineering, including writing reports and designing documents, making statement, expressing clearly, and responding instructions; Have ability to communicate in cross-cultural background and have certain international perspectives;
11. Project management: Have ability to understand and master the theory of engineering management and the method of economic decision-making, to apply them in the field of vehicle engineering under the multidisciplinary background;
12. Lifelong learning: Have consciousness of autonomous learning and lifelong learning and have ability of continuous studying and adapting to development.

附：培养目标实现矩阵

|        | 培养目标 1 | 培养目标 2 | 培养目标 3 | 培养目标 4 | 培养目标 5 | 培养目标 6 |
|--------|--------|--------|--------|--------|--------|--------|
| 毕业要求 1 |        | √      |        |        |        |        |
| 毕业要求 2 |        | √      |        |        |        |        |

|         | 培养目标 1 | 培养目标 2 | 培养目标 3 | 培养目标 4 | 培养目标 5 | 培养目标 6 |
|---------|--------|--------|--------|--------|--------|--------|
| 毕业要求 3  |        | √      | √      | √      |        |        |
| 毕业要求 4  |        | √      | √      |        |        |        |
| 毕业要求 5  |        |        | √      |        |        |        |
| 毕业要求 6  | √      |        |        | √      |        |        |
| 毕业要求 7  |        |        |        | √      |        |        |
| 毕业要求 8  | √      |        |        |        | √      |        |
| 毕业要求 9  |        |        |        |        | √      |        |
| 毕业要求 10 |        |        |        |        | √      |        |
| 毕业要求 11 |        | √      |        | √      |        |        |
| 毕业要求 12 |        |        |        |        |        | √      |

### 五、专业主干课程

现代汽车工程导论、汽车构造、汽车理论、汽车设计、汽车试验学、汽车制造工艺学、智能汽车感知技术、智能汽车决策与控制技术、汽车智能底盘技术。

### V. Core courses

Introduction to Major Course, Automobile Structure, Automobile Theory, Automobile Design, Automotive Experimental Theory, Automobile Manufacturing Technology, Intelligent Vehicle Sensing Technology, Intelligent Vehicle Decision and Control Technology, Automotive Intelligent Chassis Technology.

### 六、基本学制：四年

### VI. Recommended length of the program: 4 years

### 七、授予学位：工学学士

### VII. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

### 八、毕业学分要求：160学分

| 课程类型     |    | 学分要求 | 课程类型     |          | 学分要求 |
|----------|----|------|----------|----------|------|
| 1、公共课程平台 |    | 41   | 3、专业课程模块 |          | 42   |
| 公共基础课程   |    | 28   | 专业必修课程   |          | 28.5 |
| 通识教育课程   | 必修 | 7    | 专业选修课程   |          | 13.5 |
|          | 选修 | 6    | 4、实践教学模块 |          | 24   |
| 2、学科基础平台 |    | 47   | 专业实践课程   | 必修       | 24   |
| 专业学科基础课程 | 必修 | 43.5 |          | 选修       | 0    |
|          | 选修 | 3.5  |          | 5、素质拓展模块 |      |

\*通识教育选修课6学分包括：理工类、医学类、人文社科类、经济管理类中选择2学分（学生在本专业所属类别外的3个类中选择2个学分）；思想政治及新时代素质教育类选择2学分（“四史”1个学分，国家安全教育1个学分）；美育教育类选择2学分（美学和艺术史论类、艺术鉴赏和评论类课程至少选修1个学分）。

### VIII. Credits required for graduation: 160 credits

| Type of courses                |                  | Academic credits | Type of courses                    |                  | Academic credits |
|--------------------------------|------------------|------------------|------------------------------------|------------------|------------------|
| 1.Common Courses               |                  | 41               | 3. Specialized Courses             |                  | 42               |
| Common Basic Courses           |                  | 28               | Required Courses                   |                  | 28.5             |
| General Education Courses      | Required Courses | 7                | Elective Courses                   |                  | 13.5             |
|                                | Elective Courses | 6                | 4.Practicum and Internship Courses |                  | 24               |
| 2.General Disciplinary Courses |                  | 47               | Disciplinary Practical Courses     | Required Courses | 24               |
| Disciplinary Basic Courses     | Required Courses | 43.5             |                                    | Elective Courses | 0                |
|                                | Elective Courses | 3.5              | 5.Quality Development Courses      |                  | 6                |

### 九、学分比例

#### IX. Ratio of Credits

##### 1、必修选修学分比例

The proportion of compulsory elective credits

| 类别 | 学分  | 占总学分比例 |
|----|-----|--------|
| 必修 | 137 | 85.6%  |
| 选修 | 23  | 14.4%  |

##### 2、实践教学环节学分比例

The Proportion of credits in practice teaching

| 类型     |        | 学分     | 占总学分比例 |
|--------|--------|--------|--------|
| 实践教学环节 | 实验教学学分 | 19.125 | 30.70% |
|        | 实践教学模块 | 24     |        |
|        | 素质拓展模块 | 6      |        |

### 十、辅修

修读本专业辅修课程达到26学分且主修专业达到毕业要求者，颁发辅修专业毕业证书。修读本专业辅修课程达到40学分（含实践教学），并达到辅修学位授予条件的，颁发辅修学位证书。

#### X. Minor

The minor certificate will be issued to those who have completed 26 credits of minor courses and their own major has met the graduation requirements. The minor degree certificate will be issued to those who have completed 40 credits of minor courses (including practical teaching) and meet the conditions for awarding minor degree.

## 十一、毕业要求实现矩阵

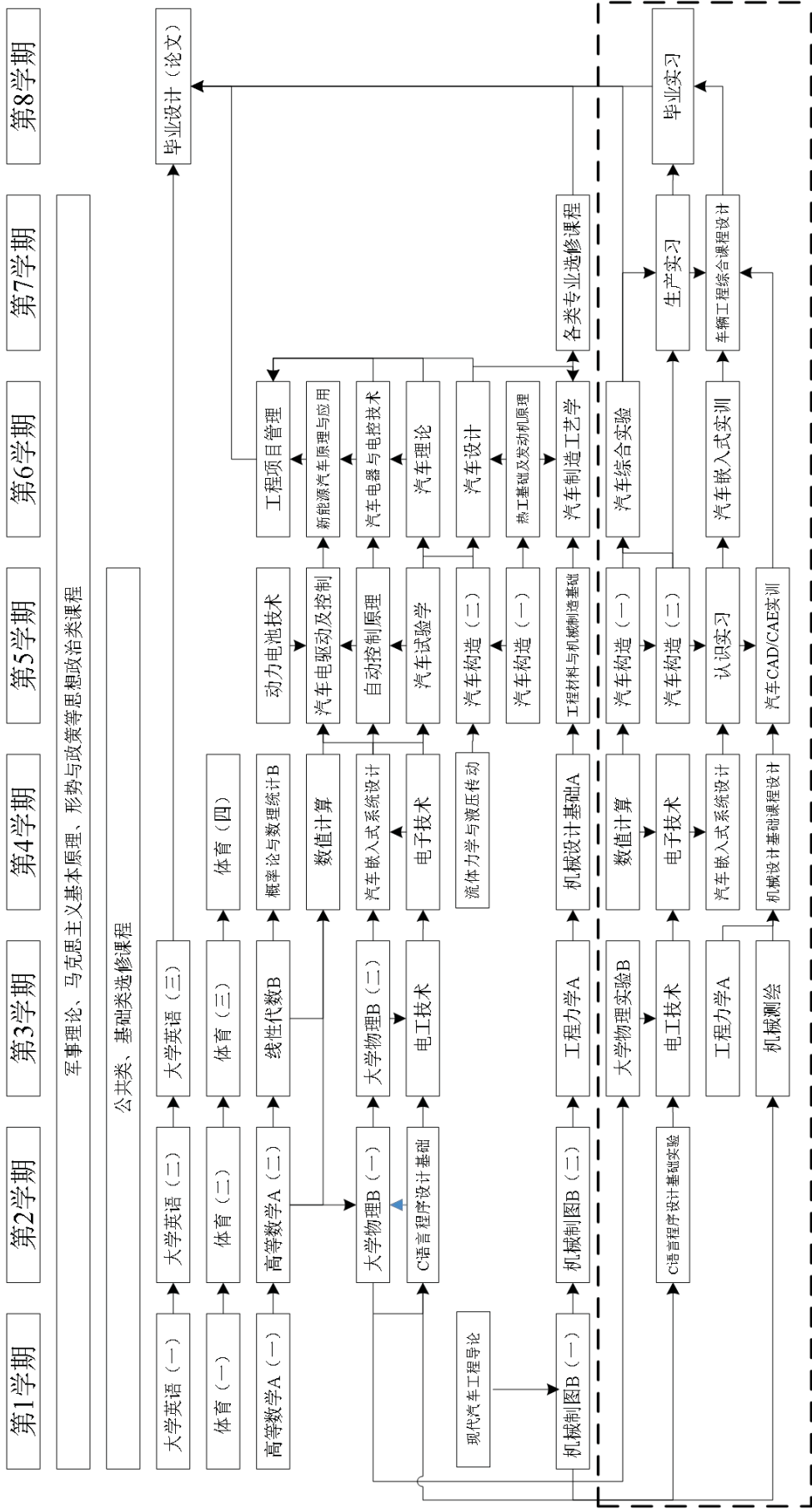
## XI. Graduation Realization Matrix

| 序号 | 课程类别     | 课程名称                 | 车辆工程专业“战略性新兴产业（支柱）产业”毕业要求 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|----|----------|----------------------|---------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|    |          |                      | 1                         |     | 2   |     | 3   |     | 4   |     | 5   |     | 6   |     | 7   | 8   |     | 9   | 10  |     | 11  | 12  |
|    |          |                      | 1.1                       | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 | 1.1 | 1.2 |
| 1  | 人文与社会科学类 | 毛泽东思想和中国特色社会主义理论体系概论 |                           |     |     |     |     |     |     |     |     |     |     |     |     | H   |     |     |     |     |     |     |
| 2  |          | 习近平新时代中国特色社会主义思想概论   |                           |     |     |     |     |     |     |     |     |     |     |     |     | H   |     |     |     |     |     |     |
| 3  |          | 思想道德与法治              |                           |     |     |     |     |     |     | M   |     |     |     |     | M   |     |     | M   |     |     |     |     |
| 4  |          | 形势与政策                |                           |     |     |     |     |     |     |     |     |     |     |     | H   | M   |     |     |     |     |     |     |
| 5  |          | 中国近现代史纲要             |                           |     |     |     |     |     |     |     |     |     |     |     |     | M   |     |     |     |     | M   |     |
| 6  |          | 马克思主义基本原理            |                           |     |     |     |     |     |     |     |     |     |     |     |     | M   |     |     |     |     |     |     |
| 7  |          | 大学英语                 |                           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     | MM  |     |     |     |
| 8  |          | 体育                   |                           |     |     |     |     |     |     |     |     |     |     |     |     |     | M   |     |     |     |     |     |
| 9  |          | 职业生涯规划与就业指导          |                           |     |     |     |     |     |     |     |     |     |     |     |     |     | M   |     |     |     | H   |     |
| 10 |          | 军事理论                 |                           |     |     |     |     |     |     |     |     |     |     |     |     | L   |     |     |     |     |     |     |
| 11 |          | 大学生创新创业基础            |                           |     |     |     |     |     |     |     |     |     |     |     |     |     | M   |     |     |     |     |     |
| 12 |          | 大学生心理健康教育            |                           |     |     |     |     |     |     |     |     |     |     |     |     |     |     | L   |     |     |     |     |
| 13 | 数学与自然科学类 | 高等数学 A               | H                         | H   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 14 |          | 概率论与数理统计 B           |                           |     |     | L   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 15 |          | 线性代数 B               | M                         | L   |     |     | L   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 16 |          | 数值计算                 |                           |     | M   |     |     |     |     |     | M   | H   |     |     |     |     |     |     |     |     |     |     |
| 17 |          | 大学物理 B               | M                         |     |     |     | L   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 18 |          | 汽车嵌入式系统设计            |                           |     | L   |     |     |     |     |     |     | M   |     |     |     |     |     |     |     |     |     |     |
| 19 |          | 大学基础化学               | L                         |     |     |     | M   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 20 | 工程基础类    | 工程力学 A               |                           | M   |     |     |     | L   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 21 |          | 电工技术                 |                           |     |     | L   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 22 |          | 电子技术                 |                           |     |     | L   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 23 |          | 自动控制原理               |                           | M   |     |     |     | M   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |

| 序号 | 课程类别   | 课程名称          | 车辆工程专业“战略性新兴产业（支柱）产业”毕业要求 |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
|----|--------|---------------|---------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--|
|    |        |               | 1                         |       | 2     |       |       |       | 3     |       |       |       | 4     |       |       |       | 5     |       | 6     | 7     | 8     |       |       |       | 9     | 10    |       | 11    | 12    |       |        |        |        |        |        |        |        |        |  |
|    |        |               | 1.1.1                     | 1.1.2 | 1.1.3 | 1.1.4 | 1.2.1 | 2.2.2 | 2.2.3 | 2.3.4 | 3.3.1 | 3.3.2 | 3.3.3 | 3.3.4 | 4.4.1 | 4.4.2 | 4.4.3 | 4.4.4 | 5.5.1 | 5.5.2 | 5.5.3 | 6.6.1 | 6.6.2 | 7.7.1 | 7.7.2 | 8.8.1 | 8.8.2 | 8.8.3 | 9.9.1 | 9.9.2 | 10.0.1 | 10.0.2 | 10.0.3 | 11.1.1 | 11.1.2 | 11.1.3 | 12.2.1 | 12.2.2 |  |
| 24 |        | 热工基础及发动机原理    | M                         |       |       |       |       | M     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 25 |        | 流体力学与液压传动     |                           |       |       | L     |       | M     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 26 | 专业基础类  | 工程材料与机械制造基础   |                           |       |       |       |       |       | M     | L     | M     |       |       |       |       |       |       |       |       |       |       | L     |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 27 |        | 机械设计基础 A      |                           |       |       | H     |       | H     |       | M     |       |       | M     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 28 |        | 智能汽车感知技术      |                           |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 29 |        | 机械制图 B        | M                         |       |       |       |       |       |       |       | L     |       |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 30 |        | 工程项目管理        |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        | M      |        |        | H      | H      |  |
| 31 | 专业类    | 汽车构造          | M                         |       |       |       |       | M     |       |       |       |       |       |       | L     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | L      |  |
| 32 |        | 汽车理论          |                           |       |       | H     | H     |       |       |       |       |       |       |       |       | M     |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 33 |        | 汽车智能底盘技术      |                           |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 34 |        | 汽车设计          |                           |       |       |       |       |       |       |       |       | H     | H     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        | L      |        |        |        |  |
| 35 |        | 汽车试验学         |                           |       |       |       |       |       |       |       |       |       |       |       |       | H     |       | M     | M     |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 36 |        | 现代汽车工程导论      |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |        |        |        | L      | H      |        |        | M      |  |
| 37 |        | 智能汽车决策与控制技术   |                           |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 38 |        | 汽车电器与电控技术     |                           |       |       |       | M     |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       | L     |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 39 | 集中实践环节 | 机械测绘          |                           |       |       |       |       |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        | L      |        |        |        |        |        |  |
| 40 |        | 大学物理实验 B      |                           |       |       |       |       |       |       |       |       |       |       |       |       | M     | M     |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 41 |        | 汽车综合实验        |                           |       |       |       |       |       |       |       |       |       |       |       |       | H     | H     | H     | H     |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 42 |        | 机械设计基础课程设计    |                           |       |       |       |       |       | H     | H     |       |       | H     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | M      |  |
| 43 |        | 工程训练 A        |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | H     |       | H     | M     |       |       |       |        |        |        |        |        |        |        |        |  |
| 44 |        | 汽车 CAD/CAE 实训 |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       | H     | H     | M     |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 45 |        | 车辆工程综合课程设计    |                           |       |       |       |       |       |       | H     |       |       | M     | M     |       |       |       |       | M     |       |       |       |       |       |       |       |       |       | H     |       | H      |        |        |        |        |        |        | M      |  |
| 46 |        | 智能汽车实训        |                           |       |       |       |       |       |       |       |       |       |       |       |       | M     |       |       |       |       | M     | L     |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |  |
| 47 |        | 创新创业实践        |                           |       |       |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |       |       |       |       |       |       | M     |       | M     |       |        |        |        |        | M      |        |        |        |  |
| 48 |        | 认识实习          |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | H     |       |       |       |       |       |       |       |        |        |        |        |        | M      |        |        |  |
| 49 |        | 生产实习          |                           |       |       |       |       |       |       |       |       | L     |       |       |       |       |       |       |       |       |       |       |       |       | M     |       |       | H     |       | H     | M      |        |        |        |        |        |        |        |  |

| 序号 | 课程类别 | 课程名称     | 车辆工程专业“战略性新兴产业（支柱）产业”毕业要求 |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |         |         |         |         |         |
|----|------|----------|---------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|--------|---------|---------|---------|---------|---------|
|    |      |          | 1                         |       | 2     |       | 3     |       | 4     |       | 5     |       | 6     |       | 7     |       | 8     |       | 9     |       | 10    |       | 11    |       | 12    |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |         |         |         |         |         |
|    |      |          | 1.1.1                     | 1.1.2 | 1.1.3 | 1.1.4 | 1.2.1 | 1.2.2 | 1.2.3 | 1.2.4 | 2.3.1 | 2.3.2 | 2.3.3 | 2.3.4 | 3.4.1 | 3.4.2 | 3.4.3 | 3.4.4 | 4.5.1 | 4.5.2 | 4.5.3 | 4.5.4 | 5.6.1 | 5.6.2 | 5.6.3 | 5.6.4 | 6.7.1 | 6.7.2 | 6.7.3 | 6.7.4 | 7.8.1 | 7.8.2 | 7.8.3 | 7.8.4 | 8.9.1 | 8.9.2 | 8.9.3 | 8.9.4 | 9.10.1 | 9.10.2 | 9.10.3 | 9.10.4 | 10.11.1 | 10.11.2 | 10.11.3 | 10.11.4 | 11.12.1 |
| 50 |      | 毕业实习     |                           |       |       |       |       |       |       | M     |       |       |       |       |       |       |       |       |       |       |       |       | M     | H     |       |       |       |       | H     | H     |       | M     |       |       |       |       |       |       |        |        |        |        |         |         |         |         |         |
| 51 |      | 毕业设计(论文) |                           |       |       |       |       |       |       |       |       | H     | H     |       |       |       |       |       |       |       |       |       | H     | H     |       |       |       |       |       |       |       |       | H     |       | H     |       | H     |       |        |        |        |        |         |         |         |         |         |
| 52 |      | 军事训练     |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | L     |       |       |       |       |       |       |       |        |        |        |        |         |         |         |         |         |
| 53 |      | 大学生劳动教育  |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | L     |       |       |       |       |       |        |        |        |        |         |         |         |         |         |
| 54 |      | 第二课堂     |                           |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | M     |       |       |       |       |        |        |        |        |         |         |         |         |         |

十二、课程修读进程表  
XII. Course review Process Map  
(四年制本科 2024级 车辆工程专业“产业计划”)



十三、教学环节设置及学分分布表

(四年制本科 2024级 车辆工程专业“产业计划”)

XIII. Offered Course and Distribution of Academic Credits

| 课 程 类 型 |          |        | 课程性质 | 课 程 编 码 | 课 程 名 称                                                                                                             | 学 分 | 合 计 | 课 内 学 时 |     |     | 实践学时 | 学 期 | 是否辅修/双学位课程 | 先 修 课 程/备 注 |
|---------|----------|--------|------|---------|---------------------------------------------------------------------------------------------------------------------|-----|-----|---------|-----|-----|------|-----|------------|-------------|
|         |          |        |      |         |                                                                                                                     |     |     | 讲 课     | 实 验 | 上 机 |      |     |            |             |
| 平台      | 公共课程平台课程 | 公共基础课程 | 必修   | 1401048 | 大学综合英语(一)<br>College English (I)                                                                                    | 2   | 32  | 32      | 0   | 0   | 0    | 1   |            |             |
|         |          |        |      | 1501882 | 体育(一)<br>Physical Education(I)                                                                                      | 1   | 26  | 26      | 0   | 0   | 0    | 1   |            |             |
|         |          |        |      | 5103001 | 中国近现代史纲要<br>An Outline of Modern and Contemporary History of China                                                  | 3   | 48  | 42      | 0   | 0   | 6    | 1   |            |             |
|         |          |        |      | 5106001 | 形势与政策<br>World Affairs and State Policy                                                                             | 2   | 64  | 64      | 0   | 0   | 0    | 1-8 |            | 分散进行        |
|         |          |        |      | 1401072 | 大学综合英语(二)<br>College English (II)                                                                                   | 2   | 32  | 32      | 0   | 0   | 0    | 2   |            |             |
|         |          |        |      | 1501883 | 体育(二)<br>Physical Education(II)                                                                                     | 1   | 34  | 34      | 0   | 0   | 0    | 2   |            |             |
|         |          |        |      | 5105017 | 思想道德与法治<br>Ideology and morality and rule of law                                                                    | 3   | 48  | 42      | 0   | 0   | 6    | 2   |            |             |
|         |          |        |      | 1401049 | 大学综合英语(三)<br>College English (III)                                                                                  | 2   | 32  | 32      | 0   | 0   | 0    | 3   |            |             |
|         |          |        |      | 1501884 | 体育(三)<br>Physical Education(III)                                                                                    | 1   | 34  | 34      | 0   | 0   | 0    | 3   |            |             |
|         |          |        |      | 5101005 | 毛泽东思想和中国特色社会主义理论体系概论<br>Theoretical System of Socialism with Chinese Characteristics                                | 3   | 48  | 32      | 0   | 0   | 16   | 3   |            |             |
|         |          |        |      | 5101006 | 习近平新时代中国特色社会主义思想概论<br>An Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era | 3   | 48  | 32      | 0   | 0   | 16   | 3   |            |             |
|         |          |        |      | 1401074 | 大学综合英语(四)<br>College English (IV)                                                                                   | 1   | 16  | 16      | 0   | 0   | 0    | 4   |            |             |
|         |          |        |      | 1501885 | 体育(四)<br>Physical Education(IV)                                                                                     | 1   | 34  | 34      | 0   | 0   | 0    | 4   |            |             |
|         |          |        |      | 5102001 | 马克思主义基本原理<br>Fundamentals of Marxism                                                                                | 3   | 48  | 44      | 0   | 0   | 4    | 4   |            |             |
|         |          |        |      | 1501024 | 体质健康测试<br>Physical Health Test                                                                                      | 0   | 16  | 0       | 0   | 0   | 16   | 1-8 |            | 分散进行        |
|         | 通识教育课程   |        | 必修   | 2502006 | 大学生心理健康教育<br>Mental Health Education                                                                                | 2   | 32  | 24      | 0   | 0   | 8    | 1   |            |             |

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| 课<br>程<br>类<br>型 | 课程<br>性质 | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                          | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------|------------------|-------------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |          |                  |                                                                                           |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平台               | 公共课程平台课程 | 必修               | 8001003 大学生创新创业基础<br>Fundamentals of Innovation and Entrepreneurship for College Students | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 2503001 职业生涯规划与就业指导<br>Career Planning and Employment Guidance                            | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 2504005 军事理论<br>Military Theory                                                           | 2      | 36     | 36      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 2501016 大学生劳动教育<br>Labor Education for College Students                                   | 1      | 32     | 0       | 0   | 0   | 32               | 1-4    |                        | 分散进行           |
|                  |          | 选修               | 美育教育类 2 学分<br>Art Education2 Academic Credit                                              |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 医学类 1 学分<br>Medicine1 Academic Credit                                                     |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 思想政治及新时代素质教育类 2 学分<br>Ideological and Political Education2 Academic Credit                |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 人文社科类 1 学分<br>Humanity and Social Sciencel Academic Credit                                |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  | 经济管理类 1 学分<br>Economic and Managementl Academic Credit                                    |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  |                                                                                           |        |        |         |     |     |                  |        |                        |                |
|                  |          |                  |                                                                                           |        |        |         |     |     |                  |        |                        |                |
|                  | 学科基础平台   | 必修               | 0302607 机械制图B(一)<br>Mechanical Drawing B(I)                                               | 2.5    | 40     | 34      | 0   | 6   | 0                | 1      |                        |                |
|                  |          |                  | 0702601 高等数学A(一)<br>Advanced Mathematics A(I)                                             | 5      | 80     | 80      | 0   | 0   | 0                | 1      |                        |                |
|                  |          |                  | 2301032 现代汽车工程导论<br>An Outline of Modern Automotive Engineering                           | 1      | 16     | 16      | 0   | 0   | 0                | 1      |                        |                |
|                  |          |                  | 0302608 机械制图B(二)<br>Mechanical Drawing B(II)                                              | 2      | 32     | 28      | 0   | 4   | 0                | 2      |                        |                |
|                  |          |                  | 0702602 高等数学A(二)<br>Advanced Mathematics A(II)                                            | 6.5    | 104    | 104     | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 0703605 大学物理B(一)<br>College Physics B(I)                                                  | 2.5    | 40     | 40      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 0401001 电工技术<br>Electrotechnics                                                           | 2      | 32     | 24      | 8   | 0   | 0                | 3      |                        |                |

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|---------|--------|----------|------|---------|----------------------------------------------------------------------|-----|-----|---------|-----|-----|------|-----|------------|-------------|
|         |        |          |      |         |                                                                      |     |     | 讲 课     | 实 验 | 上 机 |      |     |            |             |
| 平台      | 学科基础平台 | 专业学科基础课程 | 必修   | 0701605 | 工程力学A<br>Engineering Mechanics A                                     | 4.5 | 72  | 66      | 6   | 0   | 0    | 3   |            |             |
|         |        |          |      | 0702104 | 线性代数B<br>Linear Algebra B                                            | 2   | 32  | 32      | 0   | 0   | 0    | 3   |            |             |
|         |        |          |      | 0703606 | 大学物理B(二)<br>College Physics B(II)                                    | 2   | 32  | 32      | 0   | 0   | 0    | 3   |            |             |
|         |        |          |      | 0703607 | 大学物理实验B<br>Experiments of College Physics B                          | 1   | 24  | 0       | 24  | 0   | 0    | 3   |            |             |
|         |        |          |      | 0401004 | 电子技术<br>Electronic Technology                                        | 3   | 48  | 36      | 12  | 0   | 0    | 4   |            |             |
|         |        |          |      | 0702304 | 概率论与数理统计B<br>Probability and Mathematical Statistics(B)              | 2.5 | 40  | 40      | 0   | 0   | 0    | 4   |            |             |
|         |        |          |      | 2301215 | 流体力学与液压传动<br>Fluid Mechanics and Hydraulic Transmission              | 1.5 | 24  | 20      | 4   | 0   | 0    | 4   |            |             |
|         |        |          |      | 2301246 | 数值计算<br>Numerical Calculations                                       | 1.5 | 24  | 12      | 0   | 12  | 0    | 4   |            |             |
|         |        |          |      | 2301247 | 大学基础化学<br>College Basic Chemistry                                    | 1.5 | 24  | 24      | 0   | 0   | 0    | 5   |            |             |
|         |        |          |      | 2301090 | 工程项目管理<br>Engineering Project Management                             | 1   | 16  | 16      | 0   | 0   | 0    | 6   |            |             |
|         |        |          |      | 2301216 | 热工基础及发动机原理<br>Basis of Heat Energy Engineering and Engine Principles | 1.5 | 24  | 20      | 4   | 0   | 0    | 6   |            |             |
|         |        |          | 选修   | 1306009 | 大学计算机基础<br>Computer Foundation                                       | 2   | 32  | 20      | 0   | 12  | 0    | 1   |            |             |
|         |        |          |      | 1306010 | C语言程序设计基础<br>Basics of C Programming Language                        | 2.5 | 40  | 40      | 0   | 0   | 0    | 2   |            |             |
|         |        |          |      | 1306011 | C语言程序设计基础实验<br>Experiments on Basics of C Programming Language       | 1   | 24  | 0       | 0   | 24  | 0    | 2   |            |             |
|         |        |          |      | 1601004 | 信息检索与利用<br>Information Retrieval                                     | 1   | 16  | 16      | 0   | 0   | 0    | 2   |            |             |
| 模块      | 专业课程模块 | 专业必修课程   | 必修   | 0304601 | 机械设计基础A<br>Basics of Mechanical Design A                             | 4.5 | 72  | 66      | 6   | 0   | 0    | 4   |            |             |

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|---------|--------|----|---------|----------------------------------------------------------------------------------|---------|-----|-----|---------|----|----|------|-----|------------|-------------|
|         |        |    |         |                                                                                  |         |     |     | 讲课      | 实验 | 上机 |      |     |            |             |
| 模块      | 专业必修课程 | 必修 | 2301251 | 汽车嵌入式系统设计<br>Automotive Embedded System Design                                   | 2.5     | 40  | 24  | 16      | 0  | 0  | 4    |     |            |             |
|         |        |    | 2301218 | 工程材料与机械制造基础<br>Fundamentals of Engineering Material and Manufacturing Technology | 2       | 32  | 24  | 8       | 0  | 0  | 5    |     |            |             |
|         |        |    | 2301220 | 汽车构造(一)<br>Automobile Structure (I)                                              | 2       | 32  | 16  | 0       | 0  | 16 | 5    |     |            |             |
|         |        |    | 2301221 | 汽车构造(二)<br>Automobile Structure (II)                                             | 2.5     | 40  | 16  | 0       | 0  | 24 | 5    |     |            |             |
|         |        |    | 2301230 | 汽车试验学<br>Automotive Experimental Theory                                          | 1.5     | 24  | 20  | 0       | 4  | 0  | 5    |     |            |             |
|         |        |    | 2301231 | 智能汽车感知技术<br>Intelligent Vehicle Sensing Technology                               | 1.5     | 24  | 20  | 4       | 0  | 0  | 5    |     |            |             |
|         |        |    | 2301248 | 自动控制原理<br>Automatic Control System Principle                                     | 2       | 32  | 28  | 4       | 0  | 0  | 5    |     |            |             |
|         |        |    | 2301250 | 汽车智能底盘技术<br>Automotive Intelligent Chassis Technology                            | 1.5     | 24  | 24  | 0       | 0  | 0  | 5    |     |            |             |
|         |        |    | 2301012 | 汽车理论<br>Automobile Theory                                                        | 2.5     | 40  | 40  | 0       | 0  | 0  | 6    |     |            |             |
|         |        |    | 2301223 | 汽车设计<br>Automobile Design                                                        | 2       | 32  | 32  | 0       | 0  | 0  | 6    |     |            |             |
|         |        |    | 2301224 | 汽车电器与电控技术<br>Vehicle Electrical and Electronic Technology                        | 1.5     | 24  | 24  | 0       | 0  | 0  | 6    |     |            |             |
|         |        |    | 2301225 | 汽车综合实验<br>Automotive Comprehensive Experiments                                   | 1       | 24  | 0   | 24      | 0  | 0  | 6    |     |            |             |
|         |        |    | 2301232 | 智能汽车决策与控制技术<br>Intelligent Vehicle Decision and Control Technology               | 1.5     | 24  | 24  | 0       | 0  | 0  | 6    |     |            |             |
|         | 专业选修课程 | 选修 | 2301227 | 汽车电驱动及控制<br>Automotive Electric Drive and Control                                | 1.5     | 24  | 20  | 4       | 0  | 0  | 5    |     |            |             |
|         |        |    | 2301233 | 智能汽车通信技术<br>Intelligent Vehicle Communication Technology                         | 1.5     | 24  | 24  | 0       | 0  | 0  | 5    |     |            |             |

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|------------------|----------------|------------------|------------------|----------------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                |                  |                  |                |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 模块               | 专业<br>课程<br>模块 | 专业<br>选修<br>课程   | 2301234          | 汽车振动学          | 1.5    | 24      | 24  | 0   | 0                | 0      | 5                      |                |
|                  |                |                  | 2301059          | 专业英语           | 1.5    | 24      | 24  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301091          | 车身结构与设计        | 1.5    | 24      | 24  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301228          | 新能源汽车原理与应用     | 1.5    | 24      | 20  | 4   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301229          | 汽车制造工艺学        | 1.5    | 24      | 24  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301236          | 汽车检测技术         | 1.5    | 24      | 20  | 4   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301237          | 现代汽车设计方法       | 1.5    | 24      | 20  | 4   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301252          | 动力电池技术         | 1.5    | 24      | 20  | 4   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301253          | 汽车轻量化材料及应用     | 1.5    | 24      | 24  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301254          | 汽车NVH基础理论及工程分析 | 1.5    | 24      | 24  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 2301255          | 科技论文阅读与写作      | 1.5    | 24      | 24  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 2301256          | 汽车创新创业基础       | 1.5    | 24      | 24  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 2301257          | 汽车知识产权与标准      | 1.5    | 24      | 24  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 2301258          | 汽车先进技术发展前沿     | 1.5    | 24      | 24  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 2301259          | 交通系统分析         | 1.5    | 24      | 24  | 0   | 0                | 0      | 7                      |                |

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|---------|--------|--------|------|------------------------------------------------------|---------------------------------------------------------------|-----|-------|---------|-----|-----|-------|-----|------------|-------------|
|         |        |        |      |                                                      |                                                               |     |       | 讲 课     | 实 验 | 上 机 |       |     |            |             |
| 模块      | 专业课程模块 | 专业选修课程 | 选修   | 2301235                                              | 专用汽车设计<br>Special Purpose Vehicle Design                      | 1.5 | 24    | 24      | 0   | 0   | 0     | 7   |            |             |
|         |        |        |      | 1302007                                              | 人工智能导论<br>Introduction to Artificial Intelligence             | 1.5 | 24    | 24      | 0   | 0   | 0     | 7   |            |             |
|         |        |        |      | 2303067                                              | 道路交通安全<br>Road traffic Safety                                 | 1.5 | 24    | 24      | 0   | 0   | 0     | 7   |            |             |
|         | 实践教学模块 | 专业实践课程 | 必修   | 2504006                                              | 军事训练<br>Military Training                                     | 2   | 112   | 0       | 0   | 0   | 112   | 1   |            |             |
|         |        |        |      | 0302011                                              | 机械测绘<br>Mechanical Mapping                                    | 1   | 1(周)  | 0       | 0   | 0   | 1(周)  | 3   |            |             |
|         |        |        |      | 1701007                                              | 工程训练A<br>Engineering Training A                               | 2   | 4(周)  | 0       | 0   | 0   | 4(周)  | 3   |            |             |
|         |        |        |      | 0304010                                              | 机械设计基础课程设计<br>Course Project in Basics of Mechanical Design   | 1   | 2(周)  | 0       | 0   | 0   | 2(周)  | 4   |            |             |
|         |        |        |      | 2301034                                              | 认识实习<br>Introductory Practice Experience                      | 1   | 1(周)  | 0       | 0   | 0   | 1(周)  | 5   |            |             |
|         |        |        |      | 2301238                                              | 汽车CAD/CAE实训<br>Automobile CAD/CAE Training                    | 1   | 2(周)  | 0       | 0   | 0   | 2(周)  | 5   |            |             |
|         |        |        |      | 2301241                                              | 智能汽车实训<br>Intelligent Vehicle Practical Training              | 1   | 2(周)  | 0       | 0   | 0   | 2(周)  | 6   |            |             |
|         |        |        |      | 2301035                                              | 生产实习<br>Production Practice                                   | 3   | 3(周)  | 0       | 0   | 0   | 3(周)  | 7   |            |             |
|         |        |        |      | 2301239                                              | 车辆工程综合课程设计<br>Vehicle Engineering Comprehensive Course Design | 2   | 4(周)  | 0       | 0   | 0   | 4(周)  | 7   |            |             |
|         |        |        |      | 2301097                                              | 毕业实习<br>Pre-graduation Internship                             | 2   | 2(周)  | 0       | 0   | 0   | 2(周)  | 8   |            |             |
|         |        |        |      | 2301099                                              | 毕业设计(论文)<br>Undergraduate Project(Thesis)                     | 8   | 14(周) | 0       | 0   | 0   | 14(周) | 8   |            |             |
|         | 素质拓展模块 | 创新创业教育 | 必修   | 创新创业实践3学分<br>Innovation Practices 3 Academic Credits |                                                               |     |       |         |     |     |       |     |            |             |
|         |        | 第二课堂   | 必修   | 第二课堂3学分<br>Second Classroom 3 Academic Credits       |                                                               |     |       |         |     |     |       |     |            |             |

\*创新创业教育和第二课堂主要安排在小学期进行

